Machine Learning

University Textbook

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Machine Learning — Theory and Practice

Chapter 1 — Introduction to Machine Learning

Machine Learning (ML) is a subset of Artificial Intelligence that enables systems to learn from data and improve their performance over time without being explicitly programmed. Instead of writing rules manually, ML algorithms discover patterns in data and make decisions with minimal human intervention.

1.1 Types of Machine Learning

Type	Description	Example Algorithms
Supervised Learning	Learns from labelled training data to predict outputs	Linear Regression, SVM, Neural Networks
Unsupervised Learning	Finds hidden patterns in unlabelled data	K-Means, DBSCAN, PCA
Semi-Supervised	Uses a small amount of labelled data with a large amount of unlabelled data	Self-training, Label Propagation
Reinforcement Learning	Agent learns by interacting with environment and receiving rewards	Q-Learning, PPO, DDPG

1.2 The ML Pipeline

- **Step 1 — Data Collection:** Gather raw data from sources — databases, APIs, sensors, or web scraping
- **Step 2 — Data Preprocessing:** Handle missing values, remove outliers, encode categorical variables, normalize features
- **Step 3 — Exploratory Data Analysis:** Visualize distributions, correlations, and patterns using plots and statistics
- **Step 4 — Feature Engineering:** Create, select, and transform features that best represent the underlying patterns
- **Step 5 — Model Selection:** Choose appropriate algorithms based on problem type, data size, and interpretability needs
- **Step 6 — Model Training:** Fit the model on training data by optimizing a loss function
- **Step 7 — Model Evaluation:** Assess performance on held-out test data using appropriate metrics
- **Step 8 — Deployment:** Serve the trained model via API or embedded system for real-world predictions

1.3 Key Concepts

Feature (X): *An input variable used to make predictions — also called a predictor or independent variable*

Label (y): *The output variable the model tries to predict — also called the target or dependent variable*

Training Set: *Data used to fit the model parameters*

Validation Set: *Data used to tune hyperparameters and prevent overfitting*

Test Set: *Held-out data used only for final evaluation — never seen during training*

Overfitting: *Model learns training data too well including noise, performing poorly on new data*

Underfitting: *Model is too simple to capture the underlying patterns in the data*

1.4 The Bias-Variance Tradeoff

Every ML model faces the bias-variance tradeoff:

- High Bias (Underfitting): Model makes strong assumptions, fails to capture patterns. Solution: increase model complexity.
- High Variance (Overfitting): Model is too complex, memorizes training data. Solution: add regularization, get more data.
- Optimal Model: Balances bias and variance to minimize total generalization error.

1.5 Summary

Machine learning shifts the paradigm from rule-based programming to data-driven learning. The ML pipeline from data collection to deployment provides a structured framework for building predictive systems. Understanding bias-variance tradeoff and the distinction between training, validation, and test sets is essential for building models that generalize well.

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Chapter 2 — Supervised Learning Algorithms

Supervised learning is the most widely used category of machine learning. The algorithm learns a mapping function f(X) = y from labelled training examples, which it then uses to predict outputs for unseen inputs.

2.1 Linear Regression

*Linear Regression models the relationship between input features X and a continuous output y by fitting a linear equation: y = w0 + w1*x1 + w2*x2 + ... + wn*xn*

The model minimizes the Mean Squared Error (MSE) loss function using gradient descent or the normal equation:

- MSE = (1/n) * sum of (y_predicted - y_actual) squared
- Assumptions: linearity, homoscedasticity, independence of errors, no multicollinearity
- Regularization variants: Ridge (L2 penalty), Lasso (L1 penalty), ElasticNet (both)

2.2 Logistic Regression

Despite its name, Logistic Regression is a classification algorithm. It models the probability that an input belongs to a class using the sigmoid function: P(y=1|X) = 1 / (1 + e^(-z))

Aspect	Linear Regression	Logistic Regression
Output	Continuous value	Probability (0 to 1)
Loss Function	Mean Squared Error (MSE)	Binary Cross-Entropy
Use Case	Regression problems	Binary classification
Output Function	Identity	Sigmoid

2.3 Decision Trees

A Decision Tree recursively splits the data on the feature that provides the maximum information gain (or minimum Gini impurity), forming a tree of if-else rules.

- Splitting Criteria: Information Gain (ID3), Gini Impurity (CART), Gain Ratio (C4.5)
- Advantages: Interpretable, handles both numerical and categorical data, no scaling needed
- Disadvantages: Prone to overfitting; solution — pruning or ensemble methods (Random Forest)

2.4 Support Vector Machines (SVM)

SVM finds the optimal hyperplane that maximizes the margin between two classes. Points closest to the hyperplane are called support vectors.

- Linear SVM: Works when data is linearly separable
- Kernel SVM: Uses kernel functions (RBF, polynomial) to project data into higher dimensions for non-linear separation
- Hyperparameter C: Controls tradeoff between maximizing margin and minimizing classification error

2.5 Model Evaluation Metrics

Metric	Formula	Use When
Accuracy	Correct / Total	Balanced class distribution
Precision	TP / (TP + FP)	False positives are costly (spam filter)
Recall	TP / (TP + FN)	False negatives are costly (cancer detection)
F1 Score	2 * P * R / (P + R)	Need balance between precision and recall
ROC-AUC	Area under ROC curve	Comparing classifiers across thresholds
RMSE	sqrt(mean squared error)	Regression problems

2.6 Summary

Supervised learning algorithms vary widely in complexity and assumptions. Linear and logistic regression are interpretable baselines. Decision trees provide human-readable rules. SVMs excel in high-dimensional spaces. Choosing the right algorithm and evaluation metric is as important as the algorithm itself.

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Chapter 3 — Unsupervised Learning & Clustering

Unsupervised learning discovers hidden structure in unlabelled data. Without predefined output labels, these algorithms find natural groupings, reduce dimensionality, or detect anomalies purely from the input data distribution.

3.1 Clustering Overview

Clustering is the task of grouping data points such that points within the same cluster are more similar to each other than to points in other clusters.

Algorithm	Type	Key Hyperparameter	Best For
K-Means	Centroid-based	k (number of clusters)	Compact, spherical clusters
DBSCAN	Density-based	eps, min_samples	Arbitrary shapes, outlier detection
Hierarchical	Linkage-based	Linkage method	When k is unknown; dendrogram analysis
Gaussian Mixture	Probabilistic	n_components	Soft cluster assignments

3.2 K-Means Algorithm

K-Means is the most widely used clustering algorithm. It alternates between two steps until convergence:

- Step 1: Randomly initialize k centroids
- Step 2: Assign each data point to the nearest centroid (using Euclidean distance)
- Step 3: Recompute each centroid as the mean of all assigned points
- Step 4: Repeat steps 2-3 until centroids do not change significantly

Limitation: K-Means assumes spherical clusters of similar size and is sensitive to outliers. The Elbow Method and Silhouette Score help in choosing the optimal k.

3.3 Dimensionality Reduction — PCA

Principal Component Analysis (PCA) reduces the number of features by projecting data onto a lower-dimensional subspace that captures the maximum variance.

- Step 1: Standardize the data (zero mean, unit variance)
- Step 2: Compute the covariance matrix
- Step 3: Compute eigenvalues and eigenvectors
- Step 4: Select top-k eigenvectors (principal components) explaining the most variance
- Step 5: Project data onto the selected principal components

PCA is used for visualization (reduce to 2D/3D), noise removal, and as preprocessing before applying supervised learning algorithms.

3.4 Cluster Evaluation Metrics

Metric	Range	Description
Silhouette Score	-1 to +1	Measures how similar a point is to its own cluster vs ne

Davies-Bouldin Index	0 to inf	Ratio of within-cluster scatter to between-cluster separa
Inertia (WCSS)	0 to inf	Sum of squared distances from each point to its cluster
Adjusted Rand Index	-1 to +1	Measures similarity to ground truth labels (when availal

3.5 Summary

Unsupervised learning is invaluable when labels are unavailable or expensive to obtain. K-Means is simple and scalable; DBSCAN handles noise and arbitrary shapes; PCA enables visualization and feature reduction. Cluster evaluation without ground truth labels remains an open challenge — multiple metrics should be used together.